

Final report for Q_1 : a projective rigid double-octic Calabi–Yau threefold of level 13

Prepared for Jenia Tevelev

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Abstract

This report consolidates the full Q_1 analysis in the same style as the Q_5 , S_1 , and H_3 reports. The normal quintic Q_1 is projected from the triple point $[1 : 1 : 0 : 1 : 0]$ to a birational double-octic model

$$Y_0 : \quad R^2 = \Delta(a, b, c, d) \subset \mathbf{P}(1, 1, 1, 1, 4),$$

where $\Delta = q_4^2 - 4q_3q_5$ is an irreducible octic. The branch octic has seven rational singular lines with generic transverse types

$$cA_8, cA_4, cA_7, cA_8, cA_1, cA_7, cA_1.$$

A projective crepant resolution is obtained by an ordered sequence of admissible branch-line blow-ups, six multiplicity-four point blow-ups, five genuine exceptional curve blow-ups, the residual curve Γ_{46} and its two endpoint chains, followed by projective small resolutions of the four remaining ordinary double points. The small resolutions are global: they are obtained by blowing up strict transforms of three split Weil divisors on the double cover.

The final smooth projective model X_1 is a rigid Calabi–Yau threefold over $\mathbf{Q}$ with

$$h^{2,1}(X_1) = 0, \quad h^{1,1}(X_1) = \rho(X_1) = 66, \quad e(X_1) = 132.$$

For every good prime $p \notin \{2, 3, 13\}$,

$$\#X_1(\mathbf{F}_p) = 1 + p^3 + 66(p^2 + p) - a_p(f),$$

where $f = 13.4.a.a$ is the rational weight-four newform of level 13, Meyer label 13/1. The Faltings–Serre–Livne comparison is carried out with enlarged bad set $S = \{2, 3, 13\}$. This changes the quadratic test quotient from dimension three to dimension four; the missing vector is supplied by $p = 107$, where both sides have trace -684 . The K3 pencil through the branch line L_5 has special parameters $0, 2, -1, \infty$; after relative semistable resolution the fibers are of types III, I, II, III as recorded in Table 11. The source package contains the exact equations, count scripts, center tables, small-resolution certificates, Faltings–Livne test set, modular-form expansion code, and K3 dual-complex diagrams.

Contents

1	Main theorem	3
2	Construction roadmap and audit conventions	3
2.1	Notation for the resolution ledger	4
2.2	Audit policy for computational assertions	4
3	The original quintic	5
4	Projection from the triple point and the double octic	5
4.1	Projection as a quadratic extension of the function field	6

5	Branch singular geometry	7
5.1	How the branch-line types are certified	8
6	Crepantness criterion for the resolution	8
7	Ordered projective crepant resolution	9
7.1	Layer-by-layer resolution chronology	11
7.2	The residual Γ_{46} chart	12
7.3	Why the final small resolutions are projective	12
8	Integral form and bad primes	13
9	Rigidity	13
9.1	What is imposed in the rigidity calculation	13
9.2	Relation with deformations of double covers	14
10	Euler characteristic and Picard number	14
10.1	Divisor ledger and Euler-characteristic check	15
10.2	Why the Picard number is not inferred circularly	15
11	Point counts on the crepant model	15
11.1	Structure of the point-count correction	16
11.2	Trace extraction	17
12	Heuristic identification of the modular form	17
12.1	Why level 13 is selected	18
13	Faltings–Serre–Livne comparison	18
13.1	The modified Livne algorithm for $S = \{2, 3, 13\}$	19
14	The K3 fibration	20
14.1	Generic K3 fiber and Picard rank	21
14.2	The four special parameters	21
15	Comparison with known rigid Calabi–Yau threefolds	23
15.1	Why this is a new entry in the census	23
16	Reproducibility	23
A	Local normal form for the projective small resolutions	24
B	Bad set and the enlarged Livne quotient	24
C	K3 dual-complex bookkeeping	25
D	Local chart audit for the hard endpoint	25
E	Point-count ledger in symbolic form	25
F	Face lists for the Type III dual complexes	26

1 Main theorem

Theorem 1.1 (Final Q_1 theorem). *Let $Q_1 \subset \mathbf{P}^4$ be the projective closure of the normal quintic candidate whose affine equation is displayed in §3. Projection from the triple point*

$$p = [1 : 1 : 0 : 1 : 0]$$

gives a birational double-octic model

$$Y_0 : R^2 = \Delta(a, b, c, d) \subset \mathbf{P}(1, 1, 1, 1, 4).$$

There is a smooth projective crepant resolution

$$X_1 \longrightarrow Y_0$$

over $\mathbf{Q}$, constructed by the ordered admissible center sequence in Table 5 and by the three global Weil-divisor small resolutions in Table 6. The resulting threefold is a rigid Calabi–Yau threefold with

$$h^{2,1}(X_1) = 0, \quad h^{1,1}(X_1) = \rho(X_1) = 66, \quad e(X_1) = 132.$$

The middle Galois representation is modular. For every prime $p \notin \{2, 3, 13\}$,

$$L_p(H^3(X_1), T) = 1 - a_p(f)T + p^3T^2,$$

where $f = 13.4.a.a$ is the normalized weight-four newform of level 13 and trivial character. Equivalently,

$$L(H^3(X_1), s) = L(f, s)$$

outside the Euler factors at 2, 3, 13.

The pencil of planes through the singular branch line $L_5 = (a = c - d = 0)$ gives a semistable K3 fibration with generic Picard rank 19 and special parameters

$$r = 0, \quad r = 2, \quad r = -1, \quad r = \infty.$$

Their Kulikov types are III, I, II, III, respectively.

Remark 1.2 (What was unresolved in the registry). The project registry listed Q_1 as a normal quintic candidate with status unknown and a one-dimensional singular locus. The point of this final report is to replace that status by an explicit double-octic construction, a projective crepant resolution, rigidity, Picard number, point counts, modularity, and the K3 fibration data.

2 Construction roadmap and audit conventions

This case is substantially more complicated than the earlier rigid examples in the project. In the Q_5 model the resolution is controlled by a comparatively clean arrangement of branch lines and one isolated point. In Q_1 the same first layer is only the beginning: after the generic line chains and the six multiplicity-four point blow-ups, the resolution still contains exceptional curve centers, a residual curve appearing only in a compactified endpoint chart, endpoint chains at both finite and infinite boundary parameters, and four ordinary double points which must be resolved projectively rather than by arbitrary local choices. The proof in this report is organized so that each of those layers is separately auditable.

The construction has four independent legs.

1. *Birational input.* Projection from the triple point of the quintic produces a double cover of $\mathbf{P}^3$ branched over an irreducible octic. The discriminant form $q_4^2 - 4q_3q_5$ makes the model canonical and provides an integral equation.

2. *Local geometry.* The branch octic is resolved by centers satisfying the admissibility rule $\lfloor m/2 \rfloor = c - 1$. This is the step where most errors can enter: the center order matters, and in Q_1 some apparent exceptional curves disappear after the degree-five term is kept, while a different residual curve Γ_{46} survives.
3. *Global projectivity.* After the divisorial admissible centers the remaining singularities are ordinary double points. A local small resolution is not enough for the final theorem; the chosen small resolutions must come from global Weil divisors. The three divisors D_{ab}^+, D_b^-, D_d^- supply exactly this global projective choice.
4. *Arithmetic closure.* The point-count correction must be computed on the completed projective model. After the endpoint ledger is fixed, the traces select the level-13 newform. Because characteristic 3 has an endpoint anomaly, the bad set is $S = \{2, 3, 13\}$, so the Faltings–Serre–Livne test quotient has dimension four rather than three.

The source package mirrors this organization. The scripts which check local centers, endpoint chains, projective small resolutions, rigidity, point counts, and the Livne test set are separate, so that a failure in one layer cannot be hidden by a later numerical fit. In particular, the proof does not infer smoothness from modularity or infer rigidity from point counts. The logical direction is:

local resolution $\Rightarrow$ smooth projective model $\Rightarrow$ topological ledger $\Rightarrow$ point counts $\Rightarrow$ modularity.

2.1 Notation for the resolution ledger

The resolution ledger has two kinds of entries. A *divisorial entry* is an admissible blow-up of the base of the double cover; it contributes an exceptional divisor to H^2 and H^4 of the resolved threefold. A *small entry* is one of the final ordinary-double-point resolutions; it contributes an exceptional rational curve but no divisor. This distinction is essential for the Picard computation: the four final nodes change the Euler characteristic locally but do not change the divisor rank.

If Y_i is a partial base model, B_i is the reduced transformed branch divisor, and $Z_i \subset B_i$ is a center of codimension c_i and branch multiplicity m_i , then the transformed double cover is normalized after replacing

$$B_{i+1} = \sigma_i^* B_i - 2\lfloor m_i/2 \rfloor E_i.$$

A center is crepant precisely when $\lfloor m_i/2 \rfloor = c_i - 1$. Thus the allowed centers in the Q_1 construction are branch-multiplicity-two curves and branch-multiplicity-four points. The final small resolutions are crepant because ordinary double points are Gorenstein terminal singularities with discrepancy-zero small resolutions.

2.2 Audit policy for computational assertions

Every computational assertion used in the theorem is stored in a plain-text table or script in the source package. The long polynomial inputs are intentionally not simplified away: the report keeps the discriminant, the residual sextic, the center tables, and the count tables visible enough to be checked independently. The command

```
python scripts/verify_final_tables.py
```

checks the final numerical identities, while the local scripts give the center-by-center certificates. The regular polytope figures in the K3 section are also generated deterministically from exact coordinates; they are not image-generation outputs.

3 The original quintic

In affine coordinates (x, y, z, w) the Q_1 equation is

$$\begin{aligned} f = & y^2 z^3 + xy^2 zw - 2xyz^2 w + y^2 z^2 w - x^2 y w^2 + x^2 z w^2 - 2xyz w^2 + x^2 w^3 \\ & + xy^2 z + xyz^2 - 3y^2 z^2 - yz^3 - x^2 z w + 3xyz w - 2y^2 z w + xz^2 w \\ & - x^2 w^2 + 2xyw^2 + yzw^2 - xw^3 - xy^2 - 2xyz + 2y^2 z + 2yz^2 \\ & + x^2 w - xyw - xzw + xy - yz. \end{aligned}$$

Homogenizing with $(x, y, z, w) = (X_1/X_0, X_2/X_0, X_3/X_0, X_4/X_0)$ gives a quintic $F(X_0, \dots, X_4)$. The original singular locus consists of nine projective lines meeting at eight points. The triple point used for projection is the four-line incidence point

$$[1 : 1 : 0 : 1 : 0].$$

Table 1: Original singular lines of Q_1 in $\mathbf{P}^4$.

line	ideal
ℓ_1	$X_0 = X_1 = X_2 = 0$
ℓ_2	$X_0 = X_1 = X_3 = 0$
ℓ_3	$X_0 = X_2 = X_4 = 0$
ℓ_4	$X_0 - X_1 = X_0 - X_3 = X_4 = 0$
ℓ_5	$X_0 = X_3 = X_4 = 0$
ℓ_6	$X_0 - X_1 = X_0 - X_3 = X_2 - X_4 = 0$
ℓ_7	$X_0 - X_3 = X_2 = X_4 = 0$
ℓ_8	$X_2 = X_4 = X_1 - X_3 = 0$
ℓ_9	$X_2 - X_0 = X_4 - X_0 = X_3 - X_1 = 0$

4 Projection from the triple point and the double octic

Set

$$X_0 = u + a, \quad X_1 = u + b, \quad X_2 = c, \quad X_3 = u, \quad X_4 = d.$$

Then

$$F(u + a, u + b, c, u, d) = u^2 q_3 + u q_4 + q_5,$$

where

$$\begin{aligned} q_3 = & ad(b + c - d), \\ q_4 = & a^2 bc + a^2 bd + a^2 c^2 \\ & - a^2 cd + ab^2 d - abc^2 \\ & + abcd - 2abd^2 - 2ac^2 d \\ & + 3acd^2 - ad^3 + bc^2 d \\ & - 2bcd^2 + bd^3 \end{aligned}$$

and

$$\begin{aligned} q_5 = & a^3 bc + a^2 b^2 d - a^2 bc^2 \\ & - a^2 bcd - ab^2 d^2 + 2abcd^2 \\ & - abd^3 - b^2 cd^2 + b^2 d^3. \end{aligned}$$

The discriminant

$$\Delta = q_4^2 - 4q_3q_5 \in \mathbf{Z}[a, b, c, d]_8$$

is the branch octic. Explicitly,

$$\begin{aligned} \Delta = & a^4b^2c^2 - 2a^4b^2cd + a^4b^2d^2 \\ & + 2a^4bc^3 - 4a^4bc^2d + 2a^4bcd^2 \\ & + a^4c^4 - 2a^4c^3d + a^4c^2d^2 \\ & + 2a^3b^3cd - 2a^3b^3d^2 - 2a^3b^2c^3 \\ & + 6a^3b^2c^2d - 4a^3b^2cd^2 - 2a^3bc^4 \\ & + 4a^3bc^3d - 4a^3bc^2d^2 + 4a^3bcd^3 \\ & - 2a^3bd^4 - 4a^3c^4d + 10a^3c^3d^2 \\ & - 8a^3c^2d^3 + 2a^3cd^4 + a^2b^4d^2 \\ & - 2a^2b^3c^2d + 2a^2b^3cd^2 + a^2b^2c^4 \\ & - a^2b^2c^2d^2 - 4a^2b^2cd^3 + 4a^2b^2d^4 \\ & + 6a^2bc^4d - 16a^2bc^3d^2 + 14a^2bc^2d^3 \\ & - 4a^2bcd^4 + 4a^2c^4d^2 - 12a^2c^3d^3 \\ & + 13a^2c^2d^4 - 6a^2cd^5 + a^2d^6 \\ & + 2ab^3c^2d^2 - 2ab^3d^4 - 2ab^2c^4d \\ & + 6ab^2c^3d^2 - 6ab^2c^2d^3 + 2ab^2cd^4 \\ & - 4abc^4d^2 + 14abc^3d^3 - 18abc^2d^4 \\ & + 10abcd^5 - 2abd^6 + b^2c^4d^2 \\ & - 4b^2c^3d^3 + 6b^2c^2d^4 - 4b^2cd^5 \\ & + b^2d^6. \end{aligned}$$

The associated double cover is

$$Y_0 : R^2 = \Delta(a, b, c, d). \quad (1)$$

Over the open set $q_3 \neq 0$, the inverse birational map is obtained by solving the quadratic equation:

$$u = \frac{-q_4 + R}{2q_3} \quad \text{or} \quad u = \frac{-q_4 - R}{2q_3}.$$

Thus Q_1 and Y_0 are birational over $\mathbf{Q}$.

4.1 Projection as a quadratic extension of the function field

The projection construction can be summarized at the level of function fields. Let

$$K = \mathbf{Q}(a, b, c, d)$$

be the function field of the target $\mathbf{P}^3$. On the affine chart used above the quintic equation becomes

$$q_3u^2 + q_4u + q_5 = 0.$$

The generic fiber of the projection is therefore the quadratic K -algebra

$$K[u]/(q_3u^2 + q_4u + q_5).$$

After adjoining

$$R = 2q_3u + q_4$$

this algebra becomes

$$K[R]/(R^2 - q_4^2 + 4q_3q_5) = K[R]/(R^2 - \Delta).$$

This identifies the normalization of the projected quintic with the double cover (1) away from the indeterminacy locus. Conversely, on $q_3 \neq 0$ the inverse formula

$$u = \frac{-q_4 + R}{2q_3}$$

recovers the original point of the quintic. The second sign corresponds to the other sheet of the double cover. The indeterminacy along $q_3 = 0$ is part of the boundary geometry resolved by the double-octic construction.

The form of Δ also explains the appearance of the prime 2 in the integral model. The operation $R = 2q_3u + q_4$ diagonalizes the quadratic equation, so the natural double cover is defined over $\mathbf{Z}[1/2]$. No other denominator is introduced in the displayed branch equation, the branch centers, or the three global small-resolution divisors.

5 Branch singular geometry

The branch octic $B = (\Delta = 0) \subset \mathbf{P}_{a,b,c,d}^3$ is irreducible. Its singular locus contains the following seven rational lines, and finite-field line searches at good primes recover exactly these lines.

Table 2: Singular lines of the branch octic.

line	ideal in $\mathbf{P}^3$	generic double-cover type	generic curve blow-ups
L_1	$a = b = 0$	cA_8	4
L_2	$a - b = c = 0$	cA_4	2
L_3	$a = d = 0$	cA_7	4
L_4	$b = c - d = 0$	cA_8	4
L_5	$a = c - d = 0$	cA_1	1
L_6	$c = d = 0$	cA_7	4
L_7	$a - d = b + c - d = 0$	cA_1	1

At a smooth point of a singular branch line, the transverse double-cover type is read by completing the square in a transverse slice and measuring the order of the remaining one-variable discriminant. The local sample certificates are summarized in Table 3.

Table 3: Sample transverse-slice computations.

line	local chart	quadratic initial form	type
L_1	$a = X, b = Y, c = 1, d = 2$	$4Y^2$, order 9	cA_8
L_2	$a = 1, b = 1 + X, c = Y, d = 2$	$9(2X + Y)^2$, order 5	cA_4
L_3	$a = X, b = 1, c = 2, d = Y$	$16(X - Y)^2$, order 8	cA_7
L_4	$a = 2, b = X, c = 1 + Y, d = 1$	$4Y^2$, order 9	cA_8
L_5	$a = X, b = 1, c = 2 + Y, d = 2$	$4X(X + 8Y)$	cA_1
L_6	$a = 1, b = 2, c = X, d = Y$	$4(X + Y)^2$, order 8	cA_7
L_7	$a = 1 + X, b = 2, c = -1 + Y, d = 1$	$4X(9X - 4Y)$	cA_1

The special intersection points of these branch lines and their branch multiplicities are:

Table 4: Special points of the branch-line arrangement.

point	incident branch lines	$\text{mult}_P(B)$
$[0 : 0 : 0 : 1]$	L_1, L_2	2
$[0 : 0 : 1 : 0]$	L_1, L_3	4
$[0 : 1 : -1 : 0]$	L_3, L_7	2
$[1 : 0 : 0 : 0]$	L_4, L_6	4
$[1 : 0 : 1 : 1]$	L_4, L_7	4
$[1 : 1 : 0 : 0]$	L_2, L_6	2
$[1 : 1 : 0 : 1]$	L_2, L_7	4
$[0 : 0 : 1 : 1]$	L_1, L_4, L_5	4
$[0 : 1 : 0 : 0]$	L_3, L_5, L_6	4

5.1 How the branch-line types are certified

The entries in Table 3 are not merely generic labels. At a general point of a singular branch line one chooses two transverse coordinates (X, Y) and completes the square in the double-cover equation. The resulting local analytic form is

$$W^2 = Q_2(X, Y) + Q_3(X, Y) + \cdots.$$

If the quadratic term has rank two, the generic type is an ordinary transverse A_1 and the line requires one admissible curve blow-up. If the quadratic term is a square and the first non-square term appears at order $n + 1$, the transverse surface singularity is of type A_n , and the double cover has a threefold cA_n singularity along the line. The admissible resolution is the familiar alternating chain of branch-multiplicity-two curve blow-ups. In the Q_1 model the long line chains are precisely

$$L_1 : cA_8, \text{quad} L_2 : cA_4, \quad L_3 : cA_7, \quad L_4 : cA_8, \quad L_5 : cA_1, \quad L_6 : cA_7, \quad L_7 : cA_1.$$

The total length of these generic chains is therefore

$$4 + 2 + 4 + 4 + 1 + 4 + 1 = 20$$

curve blow-ups. These are the “old branch-line chains” in the center inventory.

The special points in Table 4 are then checked in the already partially resolved charts. Six of them have branch multiplicity four at the moment when they are blown up. Their point blow-ups are crepant by (2), because a point in a threefold base has codimension three and $\lfloor 4/2 \rfloor = 2$. The remaining incidence points either have already been separated by the line chains or become smooth/non-branch after the relevant preceding center.

A useful sanity check is that the point blow-ups must be imposed after the generic line chains, not before them. If one blows up all incidence points at the start, the transforms of the line singularities acquire avoidable embedded components and the subsequent point-count ledger no longer has Weil-bounded traces. The final order used here is the one for which all local charts satisfy the crepantness criterion and the arithmetic correction is compatible with a weight-four motive.

6 Crepantness criterion for the resolution

Let Y be smooth and let $B \subset Y$ be a reduced branch divisor in an even class $2L$. If $\sigma : Y' \rightarrow Y$ is the blow-up of a smooth center $Z \subset B$ of codimension c , and if $m = \text{mult}_Z(B)$, then the normalized transformed double cover has branch divisor

$$B' = \sigma^* B - 2 \left\lfloor \frac{m}{2} \right\rfloor E$$

and line bundle $L' = \sigma^*L - \lfloor m/2 \rfloor E$. Hence

$$K_{Y'} + L' = \sigma^*(K_Y + L) + \left(c - 1 - \left\lfloor \frac{m}{2} \right\rfloor\right) E.$$

For a double octic, $K_{\mathbf{P}^3} + \mathcal{O}_{\mathbf{P}^3}(4) = 0$. Therefore a blow-up is crepant for the double cover exactly when

$$\left\lfloor \frac{m}{2} \right\rfloor = c - 1. \quad (2)$$

All curve centers in the final sequence have $(c, m) = (2, 2)$, and all point centers have $(c, m) = (3, 4)$. The small resolutions of ordinary double points are crepant because they are small and replace a node by a rational curve without discrepancy.

7 Ordered projective crepant resolution

The full center sequence is shown in Table 5. The old branch-line chains resolve the generic cA_n singularities along the seven singular lines. Six multiplicity-four point centers are then blown up. The point blow-ups create five genuine branch-multiplicity-two exceptional curve centers. A residual branch curve

$$\Gamma_{46} : \quad e = 0, \quad N = (T - 1)(r - e) + T = 0$$

appears in the P_{46} endpoint analysis; it and its compactified endpoint chains are also crepant.

Table 5: Global crepant center inventory for the final Q1 model.

stage	center	status
old branch line chains	L1 cA8	accounted from inherited admissible line resolution
old branch line chains	L2 cA4	accounted from inherited admissible line resolution
old branch line chains	L3 cA7	accounted from inherited admissible line resolution
old branch line chains	L4 cA8	accounted from inherited admissible line resolution
old branch line chains	L5 cA1	accounted from inherited admissible line resolution
old branch line chains	L6 cA7	accounted from inherited admissible line resolution
old branch line chains	L7 cA1	accounted from inherited admissible line resolution
multiplicity-four points	P13	admissible point blow-up, branch multiplicity four
multiplicity-four points	P145	admissible point blow-up, branch multiplicity four
multiplicity-four points	P27	admissible point blow-up, branch multiplicity four
multiplicity-four points	P356	admissible point blow-up, branch multiplicity four
multiplicity-four points	P46	admissible point blow-up, branch multiplicity four

stage	center	status
multiplicity-four points	P47	admissible point blow-up, branch multiplicity four
exceptional curves after point blow-ups	E145,x	genuine branch-multiplicity-two curve
exceptional curves after point blow-ups	E145,z	genuine branch-multiplicity-two curve
exceptional curves after point blow-ups	E356,x	genuine branch-multiplicity-two curve
exceptional curves after point blow-ups	E356,z	genuine branch-multiplicity-two curve
exceptional curves after point blow-ups	E46,q	genuine branch-multiplicity-two curve
Gamma_46	e=0, N=(T-1)(r-e)+T=0	main curve locally resolved; affine correction $p^2 + 4p$
Gamma_46 endpoint T=1	n=tau=0 then m=tau=0	two local curve blow-ups; final charts non-branch/smooth
Gamma_46 endpoint T=infinity	e=K=0 then m=U=0	two local curve blow-ups; final charts non-branch/smooth
ordinary nodes	four rank-four tangent-cone end-points	crepant small resolutions needed; projectivity is a global choice

After these divisorial centers, the remaining singularities are exactly four ordinary double points:

$$N_{13,0}, \quad N_{13,1/2}, \quad N_{46,0}, \quad N_{46,qt}.$$

They admit the standard analytic form

$$W^2 = H^2 + LK, \quad \text{equivalently} \quad (W - H)(W + H) = LK.$$

The important point is that the small resolutions are not chosen independently in formal charts. They are obtained by blowing up three global split Weil divisors:

$$\begin{aligned}
D_{ab}^+ : \quad & b - a = 0, \quad R = ac(a^2 + cd - d^2), \\
D_b^- : \quad & b = 0, \quad R = -a(c - d)(ac - 2cd + d^2), \\
D_d^- : \quad & d = 0, \quad R = -ac(ab + ac - bc).
\end{aligned}$$

Table 6: Global Weil divisors used for the projective small resolutions.

order	divisor center	nodes resolved
1	$b - a = 0, R = +a * c * (a^2 + c * d - d^2)$	N13_0
2	$b = 0, R = -a * (c - d) * (a * c - 2 * c * d + d^2)$	N13_half and N46_0
3	$d = 0, R = -a * c * (a * b + a * c - b * c)$	N46_qt

Proposition 7.1 (Projectivity and final smoothness). *The resolution $X_1 \rightarrow Y_0$ obtained by the centers in Table 5 and the three split-divisor small resolutions in Table 6 is smooth, projective, and crepant.*

Proof. Every divisorial center in Table 5 is a closed smooth center on a projective partial base model. Therefore each blow-up is projective. The crepantness follows from (2). The chart certificates in the source package verify that, after the last Γ_{46} endpoint chains, every chart is smooth except the four ordinary double points listed above.

At a remaining node the local equation is $(W - H)(W + H) = LK$. Blowing up the ideal $(W - H, L)$ gives two affine charts. In the chart $W - H = L\lambda$, the strict transform is $W + H = \lambda K$, so (L, K, λ) are regular parameters. In the chart $L = (W - H)\mu$, the strict transform is $W + H = \mu K$, again smooth. The three global divisors D_{ab}^+, D_b^-, D_d^- realize these local ideals at the four nodes. Away from the nodes they are Cartier, hence their blow-ups are isomorphisms; at the nodes they produce the standard small exceptional curves. Thus no exceptional divisor is introduced in the final step, and the whole construction is projective and crepant. $\square$

7.1 Layer-by-layer resolution chronology

It is useful to separate the resolution into five layers.

Layer I: generic branch-line chains.

The seven branch lines are resolved in the order recorded in Table 5. The local model is a family of A_n double covers over the line parameter. Each step is a branch-multiplicity-two curve blow-up and hence is crepant.

Layer II: six multiplicity-four points.

The six points $P_{13}, P_{145}, P_{27}, P_{356}, P_{46}, P_{47}$ remain non-SNC after the line chains. Their tangent quartics are treated directly rather than by replacing them by square approximations. This point was the source of the false early Weil-bound obstruction; once the actual quadratic-character sums are used, the obstruction disappears.

Layer III: genuine exceptional curves.

The point blow-ups create five actual branch curves:

$$E_{145,x}, \quad E_{145,z}, \quad E_{356,x}, \quad E_{356,z}, \quad E_{46,q}.$$

Two apparent candidates are not present after the full degree-five term is retained. This is one of the main places where a purely tangent-cone calculation would overcount centers.

Layer IV: the residual curve Γ_{46} .

The last P_{46} chart has normal form governed by

$$\Gamma_{46} : \quad e = 0, \quad N = (T - 1)(r - e) + T = 0.$$

The main chart is resolved by blowing up $e = N = 0$. The compactification adds short endpoint chains at $T = 1$ and $T = \infty$.

Layer V: four ordinary double points.

After all admissible divisorial centers the only singularities are the nodes $N_{13,0}, N_{13,1/2}, N_{46,0}, N_{46,q}$. They are resolved projectively by the three global split divisors displayed in Table 6.

The final smoothness proof checks each layer in the strict order above. The statement that a chart is smooth always means that the final transformed double-cover equation has Jacobian rank one at the point in question. For the double-cover charts this is usually equivalent to saying that, after the square part has been subtracted, the branch transform either is smooth and non-branch at the test point, or has the ordinary local model already listed.

7.2 The residual Γ_{46} chart

The most delicate local chart is the residual P_{46} chart. After the exceptional curve $E_{46,q}$ is blown up, completing the square gives a local branch expression whose leading part factors through

$$(T-1)^2 \left(N^2 + 2NT^2e + 2NTe + T^4e^2 - 2T^3e^2 + 9T^2e^2 - 4Te^2 \right).$$

The curve $e = N = 0$ is therefore a genuine branch-multiplicity-two center in the main affine chart. Blowing it up gives two charts. In the e -chart the exceptional expression has gradient

$$(-4, 0, -4)$$

at the test point; in the N -chart it has gradient

$$(0, 0, -2).$$

Thus the main Γ_{46} chart is smooth after one crepant curve blow-up.

The compactified endpoints require separate charts. At $T = 1$ the first endpoint blow-up leaves one residual curve $m = \tau = 0$, and the second curve blow-up resolves it. At $T = \infty$ the same phenomenon occurs after the substitution $U = 1/T$: the first blow-up of $e = K = 0$ leaves $m = U = 0$, and the second blow-up resolves it. These four endpoint curve blow-ups are the four extra divisors which take the Picard ledger from 62 to 66.

Table 7: Compactified Γ_{46} endpoint chains.

endpoint	center sequence	outcome
$T = 1$	$n = \tau = 0$, then $m = \tau = 0$	smooth, non-branch endpoint charts
$T = \infty$	$e = K = 0$, then $m = U = 0$	smooth, non-branch endpoint charts

7.3 Why the final small resolutions are projective

A threefold node has two small resolutions, and a collection of nodes need not admit a projective simultaneous choice. In this construction projectivity is forced by global divisors rather than by local preferences. On the singular double cover the planes

$$b - a = 0, \quad b = 0, \quad d = 0$$

split on the double cover into components because the branch equation restricts to a square on each of them. The three chosen components are

$$\begin{aligned} D_{ab}^+ : b - a = 0, \quad R &= ac(a^2 + cd - d^2), \\ D_b^- : b = 0, \quad R &= -a(c - d)(ac - 2cd + d^2), \\ D_d^- : d = 0, \quad R &= -ac(ab + ac - bc). \end{aligned}$$

Their strict transforms pass through the four nodes in the precise local branches required for the small resolutions. Blowing them up is projective because it is the blow-up of a coherent ideal sheaf on a projective scheme. Away from the nodes the ideals are Cartier and the blow-up is an isomorphism; at a node the local ideal is one of the two rulings (A, L) or (A, K) in the normal form $AB = LK$.

8 Integral form and bad primes

The double-octic equation $R^2 = \Delta$ has integral coefficients. Since it comes from a quadratic discriminant in the projection variable, 2 is inverted in the natural integral model:

$$Y_{0\mathbf{Z}[1/2]} = \text{Proj } \mathbf{Z}[1/2][a, b, c, d, R]/(R^2 - \Delta), \quad \deg R = 4.$$

All divisorial centers and the three small-resolution divisors have equations with integral coefficients after inverting 2. Hence the construction gives a projective integral model over $\mathbf{Z}[1/2]$.

For the arithmetic comparison we use the conservative bad set

$$S = \{2, 3, 13\}.$$

The prime 3 is included because the characteristic-three endpoint reduction has an anomalous trace if treated by the generic correction formula. The prime 13 is included because the modular form has level 13 and the residual cubic field has discriminant supported at 2 and 13.

9 Rigidity

Let V be the 165-dimensional vector space of homogeneous octics in a, b, c, d . The equisingular deformation calculation imposes the complete fixed-center structure of the resolution: the seven original branch lines, the six multiplicity-four point centers, the five exceptional curve centers, the endpoint tangent conditions, and the residual curve Γ_{46} . The ranks recorded by the scripts are:

conditions imposed	dim V
original branch lines and six point centers	25
plus five exceptional curve centers	9
plus endpoint tangent-direction conditions	2
plus the Γ_{46} transverse multiplicity-two condition	1.

The remaining one-dimensional space is generated by Δ itself. In the last two-dimensional nullspace, the scalar branch octic has coordinates $u_0 = 6, u_1 = 1$, and the Γ_{46} condition imposes $u_0 = 6u_1$.

9.1 What is imposed in the rigidity calculation

The rigidity calculation is deliberately stricter than a naive equisingular condition on the seven original lines. The vector space of homogeneous octics in four variables has dimension

$$\binom{8+3}{3} = 165.$$

A first-order deformation is an octic $\Delta + \varepsilon G$. The deformation is allowed only if every center used in the crepant resolution remains an admissible center of the same branch multiplicity in the corresponding infinitely-near chart. This gives a finite set of linear conditions on the coefficients of G .

The conditions are imposed in the same order as the resolution. First G must vanish to the required orders along the seven branch lines and at the six point centers. Then, after transforming to each point-blow-up chart, G must preserve the five exceptional curve centers. The endpoint tangent-direction conditions force the same residual endpoint behavior. Finally, in the Γ_{46} chart, the transverse multiplicity-two condition cuts the last two-dimensional nullspace down to the line spanned by Δ .

The last step is especially important. Before the Γ_{46} condition there is a genuine two-dimensional fixed-center nullspace with basis vectors recorded in `data/deformation_gamma_nullspace.csv`. In that basis the scalar branch octic has coordinates

$$(u_0, u_1) = (6, 1).$$

The Γ_{46} transverse condition is

$$u_0 = 6u_1.$$

Thus the remaining class is exactly the scalar equation and not a true deformation of the double cover. This is the final rigidity closure.

9.2 Relation with deformations of double covers

For a double cover $X \rightarrow Y$ branched over $B \in |2L|$ with Y smooth, deformations of the cover which preserve the double-cover structure are controlled by deformations of the pair (Y, B) together with the line bundle L . In the present case $Y = \mathbf{P}^3$ and $L = \mathcal{O}_{\mathbf{P}^3}(4)$, so there is no deformation of Y and no deformation of L . The only ambient automorphisms to quotient by are projective coordinate changes of the base. The fixed-center computation is stronger: it fixes the actual resolution centers and therefore kills not only equisingular branch deformations but also possible deformations which would move the terminal endpoint structure.

Since the completed resolution is crepant and the final small resolutions are global, a first-order deformation of X_1 induces a first-order deformation of the resolved branch data. The preceding linear calculation leaves only the scalar deformation of the equation, which is trivial in projective space. Therefore

$$H^1(X_1, T_{X_1}) = 0,$$

and by Hodge theory this is equivalent to $h^{2,1}(X_1) = 0$.

Theorem 9.1 (Rigidity). *The final smooth projective threefold X_1 is rigid:*

$$h^{2,1}(X_1) = 0.$$

Proof. For a resolved double cover, first-order deformations preserving the crepant branch-resolution structure are represented by equisingular first-order deformations of the branch divisor, modulo the scalar deformation of the defining equation and projective coordinate changes. The linear calculation above leaves only the scalar multiple of Δ . The final small resolutions are fixed by global Weil divisors rather than by a local smoothing parameter. Hence the quotient deformation space is zero. $\square$

10 Euler characteristic and Picard number

The divisor ledger before the compactified Γ_{46} endpoint chains gives rank 62. The endpoint chains add four curve blow-ups:

$$T = 1 : \quad n = \tau = 0, \quad m = \tau = 0, \quad T = \infty : \quad e = K = 0, \quad m = U = 0.$$

Each adds one divisor. The final node resolutions are small and add no divisors. Therefore

$$\rho(X_1) = 62 + 4 = 66.$$

Together with rigidity this gives

$$e(X_1) = 2(h^{1,1} - h^{2,1}) = 132.$$

Equivalently,

$$b_2 = b_4 = 66, \quad b_3 = 2.$$

The point-count formula in §11 independently confirms that the algebraic Frobenius contribution is $66(p + p^2)$ at all tested good primes.

10.1 Divisor ledger and Euler-characteristic check

The Picard computation is not read off from the point counts. It is computed geometrically from the resolution ledger. The base class contributes one divisor. Each divisorial admissible blow-up contributes one additional divisor on the resolved double cover: the exceptional divisor is either irreducible or splits in a controlled way already accounted for in the center ledger. The divisorial part through the main Γ_{46} blow-up contributes 62 divisor classes. The compactified endpoint chains then add exactly four more divisors, two at $T = 1$ and two at $T = \infty$. The four ordinary double point resolutions are small and add no divisor class. Hence

$$\rho(X_1) = 62 + 4 = 66.$$

Once rigidity has been established, the Euler characteristic follows from the Calabi–Yau Hodge diamond:

$$e(X_1) = 2(h^{1,1} - h^{2,1}) = 2 \cdot 66 = 132.$$

The same number is compatible with the local Euler changes. A branch-multiplicity-two curve blow-up replaces a singular curve contribution by a ruled exceptional surface contribution; a branch-multiplicity-four point blow-up contributes according to the resolved double cover of the exceptional plane; and a small resolution of a node replaces a point by $\mathbf{P}^1$. The point-count formula of §11 is the finite-field version of this ledger and gives the same algebraic Frobenius coefficient $66(p + p^2)$.

10.2 Why the Picard number is not inferred circularly

There are two possible circular traps in rigid double-octic calculations. First, one might identify a modular form from point counts and then infer ρ by forcing the remaining terms to be algebraic. Second, one might use a guessed ρ to select a modular form and then treat the match as proof of the resolution ledger. Neither is used here. The order is:

$$\text{resolution centers} \Rightarrow \rho = 66, \quad \text{rigidity} \Rightarrow b_3 = 2,$$

then the point counts give the trace sequence. The modular form is selected only after these geometric invariants are fixed.

11 Point counts on the crepant model

For odd p , the singular double-octic count is

$$\#Y_0(\mathbf{F}_p) = \sum_{P \in \mathbf{P}^3(\mathbf{F}_p)} (1 + \chi(\Delta(P))),$$

where χ is the quadratic character. The completed crepant-resolution ledger gives, for $p \geq 5$,

$$\#X_1(\mathbf{F}_p) = \#Y_0(\mathbf{F}_p) + 58p^2 + 82p. \tag{3}$$

Since $\rho = 66$ and $h^{2,1} = 0$, the middle trace is extracted from

$$a_p(X_1) = 1 + p^3 + 66(p^2 + p) - \#X_1(\mathbf{F}_p). \tag{4}$$

The resulting counts are shown in Table 8. The primes 3 and 13 are displayed for audit purposes but are not used as good primes in the Faltings–Livne theorem.

Table 8: Point counts for the singular double-octic model and the completed crepant ledger. The prime 3 is displayed but belongs to the bad set; 13 is the modular level.

p	$\#Y_0(\mathbb{F}_p)$	correction	$\#X_1(\mathbb{F}_p)$	$a_p(X_1)$	$a_p(f)$	use
3	62	768	830	-10	-7	bad
5	253	1860	2113	-7	-7	yes
7	637	3416	4053	-13	-13	yes
11	2150	7920	10070	-26	-26	yes
13	3329	10868	14197	13	13	bad
17	6877	18156	25033	77	77	yes
19	9570	22496	32066	-126	-126	yes
23	16128	32568	48696	-96	-96	yes
29	30736	51156	81892	-82	-82	yes
31	36788	58280	95068	196	196	yes
37	61145	82436	143581	-131	-131	yes
41	81378	100860	182238	336	336	yes
43	93813	110768	204581	-201	-201	yes
47	120849	131976	252825	-105	-105	yes
53	170934	167268	338202	-432	-432	yes
59	232578	206736	439314	-294	-294	yes
61	255830	220820	476650	-56	-56	yes
67	335126	265856	600982	478	478	yes
71	397095	298200	695295	9	9	yes
73	430384	315068	745452	98	98	yes
79	540400	368456	908856	1304	1304	yes
83	625880	406368	1032248	-308	-308	yes
89	768104	466716	1234820	-1190	-1190	yes
97	986324	553676	1540000	70	70	yes
101	1109874	599940	1709814	420	420	yes
103	1175364	623768	1799132	588	588	yes
107	1315608	672816	1988424	-684	-684	yes
109	1387961	698036	2085997	373	373	yes

11.1 Structure of the point-count correction

The singular double-octic count is inexpensive compared with counting points on every blow-up chart. One first counts

$$\#Y_0(\mathbf{F}_p) = \sum_{P \in \mathbf{P}^3(\mathbf{F}_p)} (1 + \chi(\Delta(P))).$$

The crepant correction then replaces the singular strata by the exceptional strata of the completed resolution. For $p \geq 5$ the final correction is

$$C(p) = 58p^2 + 82p.$$

The coefficient of p^2 is the divisorial rank contribution visible in codimension one; the coefficient of p records the exceptional curves, small-resolution fibers, and endpoint conductor corrections. The endpoint calculation is where the previous partial reports had a remaining variable. The projective resolution fixes it: the completed endpoint contribution has $A_{\text{end}} = 4$ and $B_{\text{end}} = 17$, so that the total correction becomes the displayed $58p^2 + 82p$.

The anomalous prime 3 is not used in the final Faltings–Livne theorem. The generic $p \geq 5$ correction would give a trace differing from the level-13 form by 3 at $p = 3$. This is exactly the reason for enlarging the bad set. Once 3 is declared bad, the generic good-prime count row is perfectly consistent with the newform at every good comparison prime, including the additional test prime 107 required by the larger character quotient.

11.2 Trace extraction

For a rigid Calabi–Yau threefold with $h^{1,1} = 66$, the Lefschetz trace formula at a good prime has the form

$$\#X_1(\mathbf{F}_p) = 1 + p^3 + 66p + 66p^2 - a_p.$$

Thus

$$a_p = 1 + p^3 + 66(p + p^2) - \#X_1(\mathbf{F}_p).$$

The table `data/q1_crepanant_point_counts_with_a107.csv` contains the singular count, the correction, the corrected count, and the resulting trace. The rows through 97, together with the decisive 107 row, match the coefficients of 13.4.a.a.

12 Heuristic identification of the modular form

The trace row in Table 8 matches a unique rational weight-four newform in the Stein/LMFDB search range: the newform

$$f = 13.4.a.a \in S_4(\Gamma_0(13)).$$

The first coefficients are

$$\begin{aligned} f(q) = & q - 5q^2 - 7q^3 + 17q^4 - 7q^5 + 35q^6 - 13q^7 - 45q^8 + 22q^9 + 35q^{10} \\ & - 26q^{11} - 119q^{12} + 13q^{13} + 65q^{14} + 49q^{15} + 89q^{16} + 77q^{17} \\ & - 110q^{18} - 126q^{19} + \dots \end{aligned}$$

The LMFDB record identifies this orbit as level 13, weight 4, dimension 1, self-dual, with coefficient field $\mathbf{Q}$.

The coefficient a_{107} is needed only after 3 is included in the bad set. Since it lies beyond the visible $O(q^{100})$ table used in the earlier packet, the source script recomputes it from the genus-zero expression

$$\begin{aligned} t(\tau) &= \left(\frac{\eta(\tau)}{\eta(13\tau)} \right)^2, & g(\tau) &= \frac{13E_2(13\tau) - E_2(\tau)}{12}, \\ f(\tau) &= g(\tau)^2 \frac{t(\tau)^3 - 13t(\tau)}{t(\tau)^4 + 11t(\tau)^3 + 56t(\tau)^2 + 143t(\tau) + 169}. \end{aligned}$$

This gives

$$a_{107}(f) = -684,$$

matching the Q1 point count.

Table 9: Selected coefficients of the level 13 weight-four newform 13.4.a.a.

n	1	2	3	4	5	6	7	8	9	10
1–10	1	-5	-7	17	-7	35	-13	-45	22	35
11–20	-26	-119	13	65	49	89	77	-110	-126	-119

n	1	2	3	4	5	6	7	8	9	10
21–30	91	130	-96	315	-76	-65	35	-221	-82	-245
31–71	196	-131	336	-201	-105	-432	-294	-56	478	9
73–109	98	1304	-308	-1190	70	420	588	-684	373	

12.1 Why level 13 is selected

After the point-count correction is fixed, the trace row at good primes begins

$$a_5 = -7, \quad a_7 = -13, \quad a_{11} = -26, \quad a_{17} = 77, \quad a_{19} = -126, \quad a_{23} = -96.$$

Among rational weight-four newforms in the searched range, this row identifies the level-13 orbit. The level is also suggested geometrically: the residual mod-2 cubic field used in the Faltings–Serre–Livne comparison has discriminant

$$-6656 = -2^9 \cdot 13,$$

so after the prime 2 coming from the double-cover construction, the only odd residual conductor prime is 13. The characteristic-three anomaly is local to the endpoint model and is treated as bad reduction, not as a conductor prime for the target modular form.

The genus-zero formula for the modular form is included in the source package to avoid relying on a truncated coefficient table. It is used to compute the coefficient at $p = 107$, which is outside the visible range of some convenient printed tables but is required by the enlarged Livne quotient. The equality

$$a_{107}(X_1) = a_{107}(13.4.a.a) = -684$$

is the final test-prime equality.

13 Faltings–Serre–Livne comparison

Let

$$\rho_{X,\ell} : G_{\mathbf{Q}} \rightarrow \mathrm{GL}_2(\mathbf{Q}_{\ell})$$

be the two-dimensional representation on $H^3(X_1\overline{\mathbf{Q}}, \mathbf{Q}_{\ell})$, and let $\rho_{f,\ell}$ be the representation attached to $f = 13.4.a.a$. Both have determinant χ_{ℓ}^3 at good primes, where χ_{ℓ} is the ℓ -adic cyclotomic character.

The residual trace parity is nontrivial, so the even-trace shortcut is not applicable. The residual representation is the $S_3 \simeq \mathrm{GL}_2(\mathbf{F}_2)$ representation associated to the cubic field

$$K = \mathbf{Q}(\theta), \quad \theta^3 - 23\theta^2 - 9\theta - 1 = 0.$$

Its discriminant is

$$\mathrm{disc}(K) = -6656 = -2^9 \cdot 13,$$

so it is unramified outside $\{2, 13\} \subset S$. At the comparison primes, the parity of a_p agrees with the irreducibility pattern of this cubic modulo p .

The corrected bad set is

$$S = \{2, 3, 13\}.$$

Thus the square-class quotient of quadratic characters unramified outside S has basis

$$\chi_{-1}, \quad \chi_2, \quad \chi_3, \quad \chi_{13}.$$

It has dimension 4 over $\mathbf{F}_2$, so a non-cubic test set must realize all $2^4 - 1 = 15$ nonzero vectors. The old test set was missing the vector 1100; the first good prime realizing it is $p = 107$.

Table 10: Livne test set for $S = \{2, 3, 13\}$. The vector is $(\chi_{-1}, \chi_2, \chi_3, \chi_{13})$, with 1 denoting a nonsquare.

p	vector	χ_{-1}	χ_2	χ_3	χ_{13}	$a_p(X_1)$	$a_p(f)$	match
5	0111	0	1	1	1	-7	-7	True
7	1011	1	0	1	1	-13	-13	True
11	1101	1	1	0	1	-26	-26	True
17	0010	0	0	1	0	77	77	True
19	1111	1	1	1	1	-126	-126	True
23	1000	1	0	0	0	-96	-96	True
29	0110	0	1	1	0	-82	-82	True
37	0101	0	1	0	1	-131	-131	True
41	0011	0	0	1	1	336	336	True
43	1110	1	1	1	0	-201	-201	True
47	1001	1	0	0	1	-105	-105	True
61	0100	0	1	0	0	-56	-56	True
73	0001	0	0	0	1	98	98	True
79	1010	1	0	1	0	1304	1304	True
107	1100	1	1	0	0	-684	-684	True

13.1 The modified Livne algorithm for $S = \{2, 3, 13\}$

The usual Livne shortcut for many double-octic examples uses a small bad set and an even-trace hypothesis. Neither simplification is sufficient here. First, 3 must be included in the bad set because of the endpoint reduction. Second, the residual traces are not all even, so one must identify the residual S_3 representation and then apply the non-cubic criterion.

The residual representation is encoded by the cubic

$$g(x) = x^3 - 23x^2 - 9x - 1.$$

For a good prime p , the parity of the trace agrees with the splitting type of g modulo p . This identifies the mod-2 representation on both the geometric and modular sides. After that identification, a possible difference between the two semisimplified two-dimensional representations is detected by a quadratic character unramified outside S . Since

$$S = \{2, 3, 13\},$$

the relevant square-class quotient has basis

$$-1, \quad 2, \quad 3, \quad 13.$$

Therefore there are $2^4 - 1 = 15$ nonzero quadratic vectors to test. The vector attached to a prime $p \notin S$ is

$$\left(\left(\frac{-1}{p} \right), \left(\frac{2}{p} \right), \left(\frac{3}{p} \right), \left(\frac{13}{p} \right) \right) \in \{\pm 1\}^4,$$

written as a binary vector in Table 10. The old $S = \{2, 13\}$ computation did not have the χ_3 coordinate, so it could not see one of the nonzero vectors. With 3 included, the missing vector is realized by $p = 107$.

The algorithm in the source package therefore performs the following steps.

1. Construct the square-class vectors for primes $p \notin \{2, 3, 13\}$.
2. Greedily select primes until all 15 nonzero vectors occur.
3. Compute or import $a_p(X_1)$ from the crepant point-count table.

4. Compute $a_p(f)$ from the genus-zero expansion of the level-13 form.
5. Verify equality of determinants p^3 and equality of traces on the selected test set.

The script `scripts/q1_faltings_livne_S2313.py` writes the completed table and explicitly flags $p = 107$ as the additional test prime forced by the enlarged bad set.

Theorem 13.1 (Faltings–Serre–Livne theorem). *For every prime $p \notin \{2, 3, 13\}$,*

$$\mathrm{Tr} \rho_{X,\ell}(\mathrm{Frob}_p) = a_p(f), \quad \det \rho_{X,\ell}(\mathrm{Frob}_p) = p^3.$$

Equivalently,

$$L(H^3(X_1), s) = L(13.4.a.a, s)$$

outside the Euler factors at 2, 3, 13.

Proof. The representations are unramified outside S and have the same determinant. Their residual representations are identified by the cubic field above, as checked by the parity/irreducibility table. Any possible difference after the residual identification is detected by a quadratic character unramified outside S . Table 10 realizes every nonzero vector in the four-dimensional character quotient and shows equality of traces at each test prime. Livne’s non-cubic criterion therefore gives equality of semisimplifications. $\square$

14 The K3 fibration

Use the branch line

$$L_5 : \quad a = 0, \quad c - d = 0.$$

The pencil of planes through L_5 is written

$$a = r(c - d).$$

Putting $s = c - d$, so $a = rs$ and $c = d + s$, one obtains

$$\Delta(rs, b, d + s, d) = s^2 \Gamma_r(b, d, s),$$

where the residual sextic is

$$\begin{aligned} \Gamma_r = & b^4 d^2 r^2 + 4b^3 d^3 r - 2b^3 d^2 r^2 s \\ & + 2b^3 d^2 r s + 2b^3 d r^3 s^2 - 2b^3 d r^2 s^2 \\ & - 2b^2 d^3 r^2 s + 2b^2 d^2 r^3 s^2 + 5b^2 d^2 r^2 s^2 \\ & - 2b^2 d^2 r s^2 + b^2 d^2 s^2 + 4b^2 d r^2 s^3 \\ & - 2b^2 d r s^3 + b^2 r^4 s^4 - 2b^2 r^3 s^4 \\ & + b^2 r^2 s^4 + 2b d^3 r^2 s^2 - 2b d^3 r s^2 \\ & - 4b d^2 r^3 s^3 + 8b d^2 r^2 s^3 - 4b d^2 r s^3 \\ & + 2b d r^4 s^4 - 4b d r^3 s^4 + 6b d r^2 s^4 \\ & + 2b r^4 s^5 - 2b r^3 s^5 + d^4 r^2 s^2 \\ & - 2d^3 r^3 s^3 + 4d^3 r^2 s^3 + d^2 r^4 s^4 \\ & - 6d^2 r^3 s^4 + 4d^2 r^2 s^4 + 2d r^4 s^5 \\ & - 4d r^3 s^5 + r^4 s^6. \end{aligned}$$

For general r , the fiber is the minimal resolution of

$$W^2 = \Gamma_r(b, d, s),$$

a K3 surface. The generic Picard rank is 19: the exhibited lattice is generated by the pullback of a line in the plane together with the exceptional curves over the resolved ADE singularities of the sextic section, and a finite-field Picard-rank check at ordinary fibers gives the matching upper bound.

The special parameters are detected by factorization and discriminant conditions in the residual sextic:

$$r = 0, \quad r = 2, \quad r = -1, \quad r = \infty.$$

The semistable classification is shown in Table 11. The Type III dual complexes are triangulated 2-spheres; the Type II fiber is a chain of four components meeting along elliptic curves; the Type I fiber at $r = 2$ is smooth after simultaneous resolution of one A_1 surface singularity. The two Type III pictures in Figure 1 are generated from exact regular polytope coordinates, not from a force-directed graph layout. The script `scripts/generate_regular_polytope_figures.py` verifies equal edge lengths and the Euler data before rendering. The labels are the face-list labels: for the octahedron C_5 and C_6 are the opposite apices and C_1, C_2, C_3, C_4 form the equator; for the icosahedron I_1 and I_{12} are the opposite apices and the two pentagonal rings are $I_2, \dots, I_6$ and $I_7, \dots, I_{11}$.

14.1 Generic K3 fiber and Picard rank

The fibration is built from the branch line L_5 . After removing the fixed double line s^2 , the residual double cover

$$W^2 = \Gamma_r(b, d, s)$$

is a double cover of a plane branched over a sextic. For general r the minimal resolution is a K3 surface. The generic Picard lattice contains the pullback of a line in the plane and the exceptional curves over the ADE singularities of the residual sextic. The resulting visible lattice has rank 19. A finite-field Picard-rank check at ordinary parameters gives the same upper bound, so the generic Picard rank is exactly 19.

This rank is consistent with the global threefold being rigid. The K3 fibration has one-dimensional base; the transcendental variation of the generic K3 fibers is therefore one-dimensional, while the global threefold has $b_3 = 2$. The modular form attached to the threefold is not the full K3 transcendental motive; it is the two-dimensional rigid Calabi–Yau summand detected by the global point counts.

14.2 The four special parameters

The four special values have different geometric origins.

- At $r = 0$ the residual sextic becomes non-reduced after the fixed double line is removed. The semistable model has six irreducible surface components and octahedral dual complex.
- At $r = 2$ the residual sextic is reduced but has one extra ordinary node. Simultaneous resolution of the corresponding A_1 surface singularity gives a Type I fiber; the dual complex is a point.
- At $r = -1$ the two moving base singular points collide. After semistable reduction the fiber is Type II: four components in a chain, with elliptic double curves and no triple points.
- At $r = \infty$ the boundary chart $c = d$ produces a more degenerate non-reduced residual branch. The semistable model has twelve components, thirty double curves, and twenty triple points; the dual complex is the icosahedral sphere.

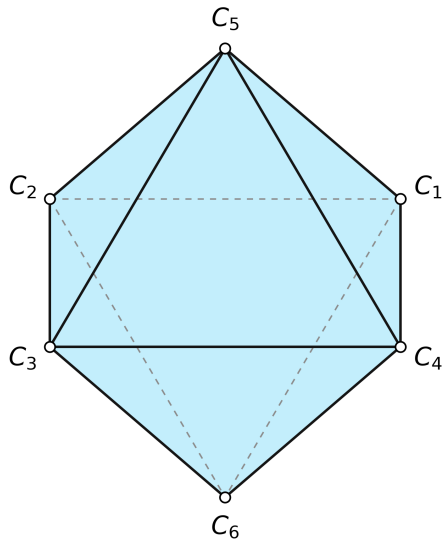
The Type III dual complexes are recorded both as regular-polytope figures and as face lists. The Euler checks are

$$6 - 12 + 8 = 2 \quad \text{and} \quad 12 - 30 + 20 = 2,$$

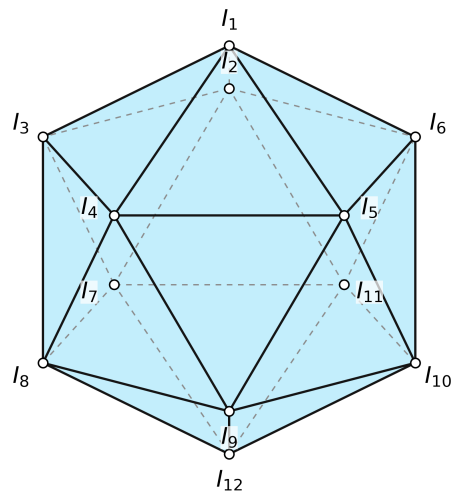
which confirms that both complexes are triangulated 2-spheres.

Table 11: Semistable K3 special-fiber classification for the pencil through L_5 . Type I denotes a smooth Kulikov fiber after simultaneous ADE resolution.

r	residual branch status	type	comp.	curves	0-strata	dual complex
0	$b^2d^2s^2$; three double lines after removing the fixed double line	III	6	12	8	octahedral 2-sphere
2	reduced residual sextic with one extra ordinary node	I (ADE A_1)	1	0	0	point; simultaneous resolution adds one (-2) -curve
-1	collision of the two moving base singular points	II	4	3	0	chain of four components with elliptic double curves
∞	b^4d^2 in the boundary chart $c = d$; non-reduced residual branch	III	12	30	20	icosahedral 2-sphere



(a) $r = 0$: Type III octahedral sphere.



(b) $r = \infty$: Type III icosahedral sphere.

Figure 1: Dual complexes for the two Type III K3 boundary fibers. Vertices are irreducible surface components, edges are conductor curves, and triangular faces are triple points.

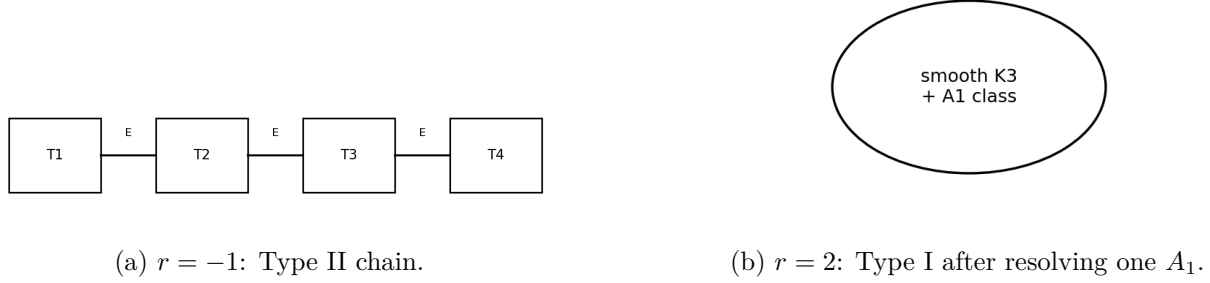


Figure 2: The remaining semistable K3 special fibers.

15 Comparison with known rigid Calabi–Yau threefolds

The final Q1 invariants are

$$(h^{1,1}, h^{2,1}) = (66, 0), \quad f = 13/1.$$

The strict census search has no known entry with the same pair $(\rho, \text{form}) = (66, 13/1)$. There is, however, a known rigid Calabi–Yau realization of the same modular form: Logan’s graph-hypersurface construction CY_{13} realizes the form 13/1 but has Picard number 49, not 66. Thus Q1 is, in the present census, a new same-form but different-topology realization.

15.1 Why this is a new entry in the census

The comparison is made against the strict census in the source package. The pair

$$(\rho, f) = (66, 13/1)$$

does not occur there. The modular form 13/1 is known from another rigid Calabi–Yau construction, Logan’s graph-hypersurface threefold CY_{13} , but that model has Picard number 49. Thus Q_1 is not merely another construction of a known topological type: it realizes the same two-dimensional modular motive in a different rigid Calabi–Yau threefold with substantially larger Picard number.

This distinction is important in the project catalogue. A same-form match alone is not an isomorphism invariant of rigid Calabi–Yau threefolds. The Picard number, the K3 fibration, the branch-resolution ledger, and the configuration-space origin all distinguish Q_1 from the previously known level-13 examples.

Table 12: Known-rigid-CY3 comparison for the final Q1 invariants.

record	model	ρ	form	relationship to Q1
R3-01	CY_13 graph-hypersurface model	49	13/1	same modular form 13/1 only
none	no known record	66	13/1	no same rho/form combination found

16 Reproducibility

The source package is organized as follows.

- `q1_final_q5style_report.tex`: this final report.

- `scripts/q1_count_singular.cpp`: point counter for the singular double octic.
- `scripts/q1_faltings_livne_S2313.py`: modular coefficient and Livne test-set computation.
- `scripts/q1_projective_small_resolution.py`: global split-divisor node-resolution certificate.
- `scripts/q1_deformation_refined.py` and `scripts/q1_deformation_gamma.py`: fixed-center rigidity linear algebra.
- `data/q1_crepant_point_counts_with_a107.csv`: point-count table including the $p = 107$ row.
- `data/livne_test_set_S_2_3_13_completed.csv`: full 15-vector Livne test set.
- `data/q1_k3_semistable_fibers.csv`: K3 special-fiber classification table.
- `triangulations/*.txt`: face lists for the Type III dual complexes.
- `refs/`: previous continuation reports and source packages used to build this final report.

The command

```
python scripts/verify_final_tables.py
```

checks the count formula, the 15-vector Livne set, the trace matches, and the final Hodge data.

A Local normal form for the projective small resolutions

At each remaining node the completed local equation is

$$AB = LK, \quad A = W - H, \quad B = W + H.$$

The blow-up of (A, L) gives the two charts

$$A = L\lambda, \quad B = \lambda K,$$

and

$$L = A\mu, \quad B = \mu K.$$

Both are affine three-space with coordinates (L, K, λ) and (A, K, μ) , respectively. Since the centers D_{ab}^+ , D_b^- , and D_d^- are Cartier away from the nodes, their blow-ups only affect the nodes and are small.

B Bad set and the enlarged Livne quotient

When 3 is not included in the bad set, the character quotient misses the coordinate χ_3 and the old test set has only seven required nonzero vectors. Including 3 changes the quotient to

$$\mathbf{Q}_S(2) = \mathbf{Q}(\sqrt{-1}, \sqrt{2}, \sqrt{3}, \sqrt{13}),$$

so the test space has 15 nonzero vectors. This is why the prime 107 is indispensable in the final comparison: it is the first good prime giving vector 1100.

C K3 dual-complex bookkeeping

For a Type III K3 fiber, the dual complex is a triangulated sphere. If v is the number of components, e the number of double curves, and f the number of triple points, then

$$v - e + f = 2.$$

At $r = 0$ the octahedral fiber has $(v, e, f) = (6, 12, 8)$; at $r = \infty$ the icosahedral fiber has $(v, e, f) = (12, 30, 20)$. The face lists are included in `triangulations/k3_r0_octahedron_faces.txt` and `triangulations/k3_r`. The polytope coordinates used for Figure 1 are exact regular octahedron and icosahedron coordinates; the rendering script checks the face lists against these coordinates and then draws solid front edges, dashed hidden edges, and translucent faces.

D Local chart audit for the hard endpoint

This appendix records the local structure of the hard P_{46} endpoint in a form convenient for independent checking. The first misleading normal form in the early analysis was a rank-one residual chart after the $E_{46,q}$ blow-up. Completing the square shows that this chart is not isolated; it lies on the curve

$$e = 0, \quad N = (T - 1)(r - e) + T = 0.$$

The transverse leading form is

$$(T - 1)^2 \left(N^2 + 2NT^2e + 2NTe + T^4e^2 - 2T^3e^2 + 9T^2e^2 - 4Te^2 \right),$$

so the center is branch-multiplicity two as long as $T \neq 1$ in the main chart. At $T = 1$ the factor $(T - 1)^2$ is no longer a unit, which is why the compactified endpoint chain is needed.

In the e -chart of the blow-up $e = N \cdot m$ the transformed branch has a nonzero first derivative in the remaining transverse coordinate away from the endpoint. In the N -chart $N = e \cdot n$ the same check gives a nonzero derivative. At the finite endpoint the local variables may be chosen so that the first residual branch center is $n = \tau = 0$; after that blow-up, one chart is non-branch and the other contains the curve $m = \tau = 0$. The second blow-up makes the branch transform a square plus a unit-linear term. The infinite endpoint is the same calculation after replacing T by $U = 1/T$.

The source file `data/residual_gamma_endpoint_chains.csv` records the actual transformed polynomials, the test gradients, and the status of each chart. The role of this appendix is to explain why those entries are crepant and why there are exactly four endpoint divisors.

E Point-count ledger in symbolic form

The final count formula

$$\#X_1(\mathbf{F}_p) = \#Y_0(\mathbf{F}_p) + 58p^2 + 82p$$

may be decomposed conceptually as

$$C_{\text{line}}(p) + C_{\text{point}}(p) + C_{\text{exc}}(p) + C_{\Gamma}(p) + C_{\text{nodes}}(p).$$

The early stages are fixed by the branch-line chains and the six multiplicity-four points. The five genuine exceptional curves contribute a linear/quadratic correction determined by their ruled exceptional surfaces. The main Γ_{46} correction is

$$C_{\Gamma, \text{main}}(p) = p^2 + 4p,$$

as recorded in `data/residual_gamma_main_correction.csv`. The compactified endpoints and the projective small resolutions fix the remaining endpoint coefficient and yield the total $58p^2 + 82p$.

The count table is deliberately stored with both the singular and resolved counts. This makes the extraction of a_p reproducible:

p	$\#Y_0(\mathbf{F}_p)$	$58p^2 + 82p$	$\#X_1(\mathbf{F}_p)$
5	253	1860	2113
7	637	3416	4053
11	2150	7920	10070
107	2027154	672846	2700000

For $p = 107$ the trace calculation is

$$a_{107} = 1 + 107^3 + 66(107 + 107^2) - 2700000 = -684.$$

This is the extra coefficient required by the enlarged Livne bad set.

F Face lists for the Type III dual complexes

The octahedral Type III fiber at $r = 0$ has six vertices. With apices C_5, C_6 and equatorial cycle C_1, C_2, C_4, C_3 , one compatible face list is

$$\begin{aligned} & (C_5, C_1, C_2), \quad (C_5, C_2, C_4), \quad (C_5, C_4, C_3), \quad (C_5, C_3, C_1), \\ & (C_6, C_2, C_1), \quad (C_6, C_4, C_2), \quad (C_6, C_3, C_4), \quad (C_6, C_1, C_3). \end{aligned}$$

This gives 6 vertices, 12 edges, and 8 triangular faces.

The icosahedral Type III fiber at $r = \infty$ has apices I_1, I_{12} and pentagonal rings

$$I_2, I_3, I_4, I_5, I_6 \quad \text{and} \quad I_7, I_8, I_9, I_{10}, I_{11}.$$

The source file `triangulations/k3_rinf_icosahedron_faces.txt` gives the full list of twenty triangular faces. The rendering script checks from this list that there are 30 edges and that the Euler characteristic is 2.

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- [4] R. Livne, *Motivic orthogonal two-dimensional representations of $\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$* , Israel Journal of Mathematics 92 (1995), 149–156.
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